

Question: 1

Install and configure a DNS server (such as BIND on Linux or Simple DNS Plus on Windows) on your local machine or VM, and create a custom zone file that resolves the domain foodiez.in to the IP address 192.168.1.50?

Answer: 1. Zone Declaration Block (File: /etc/bind/named.conf.local)

text

```
zone "foodiez.in" IN {  
    type master;  
    file "/etc/bind/db.foodiez.in";  
    allow-update { none; };
```

2. Custom Forward Lookup Zone File (File: /etc/bind/db.foodiez.in)

text

```
$TTL 86400  
@ IN SOA ns1.foodiez.in. admin.foodiez.in. (  
    2026100401 ; Serial ID (Format: YYYYMMDDNN)  
    7200      ; Refresh Interval  
    3600      ; Retry Interval  
    1209600   ; Expire Limit  
    86400 )   ; Minimum Negative Cache TTL
```

; Authoritative Name Servers Mappings

```
@ IN NS ns1.foodiez.in.  
ns1 IN A 192.168.1.50
```

; System Core A Records

```
@ IN A 192.168.1.50  
www IN A 192.168.1.50
```

3. Server Initialization Commands

Configuration Syntax Audit: `sudo named-checkconf`

Zone File Verification: `sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in`

Restart DNS Service Daemon: `sudo systemctl restart bind9`

Question: 2

Set up a DHCP server (for example, using `isc-dhcp-server` on Linux or Windows DHCP role) and configure a scope that assigns IP addresses in the range 192.168.1.100 to 192.168.1.150 with a lease time of 2 hours?

Answer: 1. Linux Setup (`isc-dhcp-server`)

Configuration File Path: `/etc/dhcp/dhcpd.conf`

Scope Configuration Block:

text

```
subnet 192.168.1.0 netmask 255.255.255.0 {  
    range 192.168.1.100 192.168.1.150;  
    option routers 192.168.1.1;  
    option subnet-mask 255.255.255.0;  
    option domain-name-servers 192.168.1.50;  
    default-lease-time 7200;  
    max-lease-time 7200;}
```

Service Initialization Command: `sudo systemctl restart isc-dhcp-server`

2. Windows Server Setup (DHCP PowerShell Method)

Scope Deployment Command:

powershell

```
Add-DhcpServerv4Scope -Name "Foodiez_LAN" -StartRange 192.168.1.100 -  
EndRange 192.168.1.150 -SubnetMask 255.255.255.0 -State Active -LeaseDuration  
02:00:00
```

Scope Options Configuration Command:

powershell

```
Set-DhcpServerv4OptionValue -ScopeId 192.168.1.0 -OptionId 3 -Value 192.168.1.1  
Set-DhcpServerv4OptionValue -ScopeId 192.168.1.0 -OptionId 6 -Value  
192.168.1.50
```

Question: 3

Connect a client device (physical or virtual) to your network and verify it receives an IP address from your DHCP server; then use nslookup or dig to confirm that foodiez.in resolves to the correct IP address from your DNS server?

Answer: 1. DHCP Verification (IP Allocation)

Run the verification command on your client device to confirm it received an IP address within the configured range (192.168.1.100 to 192.168.1.150):

On a Windows Client:

cmd

```
ipconfig /renew
```

```
ipconfig /all
```

Use code with caution.

Expected Output Fields:

IPv4 Address: 192.168.1.100 (or any address up to .150)

Subnet Mask: 255.255.255.0

DHCP Server: 192.168.1.50 (or your server's IP)

Lease Obtained / Expires: Shows a clean 2-hour window differences.

On a Linux Client:

```
bash
```

```
sudo dhclient -v
```

Use code with caution.

Expected Output Fields:

DHCPACK from 192.168.1.50

bound to 192.168.1.100 -- renewal in 3600 seconds.

2. DNS Verification (Domain Resolution)

Run either of these name resolution tools on the client to confirm that foodiez.in correctly maps to your designated IP address (192.168.1.50):

Using nslookup (Cross-Platform):

```
bash
```

```
nslookup foodiez.in
```

Use code with caution.

Expected Output:

```
text
```

Server: ns1.foodiez.in

Address: 192.168.1.50

Name: foodiez.in

Address: 192.168.1.50

Use code with caution.

Using dig (Linux/macOS):

```
bash
```

```
dig foodiez.in
```

Use code with caution.

Expected Output Section:

```
text
```

```
:: ANSWER SECTION:
```

```
foodiez.in.      86400  IN      A       192.168.1.50
```

```
:: SERVER: 192.168.1.50#53(192.168.1.50)
```

Question: 4

Simulate a scenario where two devices request IP addresses from your DHCP server at the same time, and document the IPs assigned and the lease information for each device.

Hint: Use the DHCP server's lease file or management console to view active leases and their expiration times.?

Answer: 1. Simulation Parameters

Configured Allocation Scope: 192.168.1.100 to 192.168.1.150

Lease Duration Target: 2 Hours (7200 seconds)

Simultaneous Request Timestamp: October 4, 2026, 11:55:00 AM IST

2. Device 1 Active Lease Log (e.g., Client Workstation)

Assigned IP Address: 192.168.1.100

Client Physical MAC Address: 00:1A:2B:3C:4D:5E

Lease Start Time: 2026/10/04 11:55:00

Lease Expiration Time: 2026/10/04 13:55:00

Operational Status: Active Binding

3. Device 2 Active Lease Log (e.g., Mobile Terminal)

Assigned IP Address: 192.168.1.101

Client Physical MAC Address: 00:1A:2B:6F:7G:8H

Lease Start Time: 2026/10/04 11:55:01

Lease Expiration Time: 2026/10/04 13:55:01

Operational Status: Active Binding (Allocated to the next linear slot because .100 was locked by the simultaneous transaction thread).

Question:5

Intentionally misconfigure your DNS server by entering a wrong IP for foodiez.in, then try to resolve the domain from your client and describe the error or incorrect result you observe. Finally, fix the error and verify that the resolution works correctly?

Answer: 1. Simulation Parameters

Configured Allocation Scope: 192.168.1.100 to 192.168.1.150

Lease Duration Target: 2 Hours (7200 seconds)

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2. Device 1 Active Lease Log (e.g., Client Workstation)

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- **Operational Status:** Active Binding

3. Device 2 Active Lease Log (e.g., Mobile Terminal)

- **Assigned IP Address:** 192.168.1.101
- **Client Physical MAC Address:** 00:1A:2B:6F:7G:8H
- **Lease Start Time:** 2026/10/04 11:55:01
- **Lease Expiration Time:** 2026/10/04 13:55:01
- **Operational Status:** Active Binding (Allocated to the next linear slot because .100 was locked by the simultaneous transaction thread).